Safeguard

a place to live





"At the end of the day, if you want the sky of Singapore to be always blue, the land of Singapore to be always green, the water to be always clean, the public health to be always pink, it counts a lot on these many faceless heroes in MEWR, NEA, PUB as well as many faceless heroes in the industry and the workers in the environment sector. We have good people in MEWR and NEA. They know their jobs well, and they know the challenges we face. They are passionate and committed to meeting these challenges. Most of all, they are prepared to adapt to changing circumstances so as to meet new challenges with new mindsets."

— Lim Swee Say, Minister for the Environment (2002 - 2004)



Top: Chinatown in 1970. Scenes such as this were common in the past. (*Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.*)

Opposite: Pagoda Street in 1972. (*Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.)*

Snapshot. Singapore in the 1960s. Streets were more often than not cluttered with all kinds of refuse. One could see discarded tyres, bottles, nets and unidentifiable waste floating and choking up streams, canals and rivers. The water would be black, and stenches were often released. Right next to sawmills, workshops and factories, residences would be found, and people would live amidst the dust, noise and unpleasant smells.

In a matter of decades, a complete transformation had taken place. Singapore had become clean. So much so that it is something we hardly think about today.

Clean air, clean water, clean land. These do not come by accident. When Singapore gained self-governance in 1959, it inherited a heavily polluted environment, especially in the core of the city. In the outskirts of Singapore, there was severe land pollution following decades of dumping untreated refuse in swamps, coupled with liberal use of pesticides such as DDT to prevent mosquito breeding.

At the same time the new government grappled with urgent issues such as housing needs, unemployment and high crime rates, it also began to look into eradicating and controlling pollution through appropriate planning and zoning of land use on the island. With independence in 1965, the government had greater control over the master planning of Singapore's development. Alongside an aggressive agenda for economic development and industrialisation, the government had the foresight to lay down policies ensuring that our environment would not be compromised. This commitment to protect and safeguard our living surroundings is a cornerstone of Singapore's transformation.

Such a commitment is all the more crucial in the light of the challenges we face today. With rapid globalisation and technological advancements in industrial processes, the scale of environmental incidents is potentially further-reaching and more complex than before. Increasing industrialisation also means a growing amount of emissions from our industries. Within Singapore, the environment faces more stress as the population grows and land use becomes more intensive.



The task of bringing about clean air, clean water and clean land falls on NEA, which translates the policies and direction set by MEWR into action. NEA shapes the planning of Singapore's land use, regulates the development of industries and makes sure that there is a balance between economic growth and environmental well-being. It stays on the alert and constantly monitors the quality of our air, water and land. It implements a stringent routine of licensing and enforcement to prevent environmental hazards from taking place. It also goes beyond its enforcer role to reach out to industries to educate, inform and encourage innovation.

This will not be possible without the people in the agency, who work to safeguard and protect the environment. Much of what they do is not glamorous, and for some, it is tedious. Yet talk to any of the workers and they often speak of their work with pride. They are the people who bring industries in line with our strict requirements, who go to the ground to iron out issues. They are planners, inspectors, incineration plant operators, technicians, scientists and engineers, working in concert to keep our home clean and green. A home where we can breathe in clean air, enjoy clean beaches and work in liveable, sustainable surroundings.

"The transition of the operations of environmental protection from the ministry to NEA in 2002 was smooth as the planning had been thorough and comprehensive. Nonetheless, there was a sense of autonomy from the ministry and with that came greater responsibility and accountability. We worked to ensure that Singaporeans could continue to enjoy clean air, clean water, clean land and a high standard of public health, and this was realised with the strong commitment, conviction and the camaraderie of the officers in NEA working to protect the environment."

— Loh Ah Tuan, Director-General, Environmental Protection Division, NEA (2002 - 2007)





Safeguarding Essentials

Sound principles, a strong legal framework, thorough planning. Add to that a rigorous regime of surveillance, licensing and enforcement, constant innovation and the roping in of different stakeholders. Such is the formula employed by NEA to safeguard Singapore's environment, to make sure that our air, water, and land are clean.

THE PRINCIPLES OF IT ALL

Pollution that goes unchecked is one of the greatest ills of industrialisation.

Take, for example, the case of widespread mercury poisoning during the 1950s in Minamata, Japan, after a chemical factory released industrial wastewater with high mercury content into the city's waterways. Or the discovery in the 1950s and 1960s that the use of leaded petrol was linked to lead poisoning in humans and lower intelligence in children.

There are many other similar cases. Singapore, being a relatively newcomer to industrialisation, has the advantage of learning from such experiences in other countries. Which is how it formed three simple principles for environmental protection.

The first principle is to exercise "pollution control at source". Literally, this means controlling pollution right where it happens before it escalates into bigger issues. Key sources of pollution – industrial facilities, power plants and vehicles – are regulated to make sure they do not pollute the environment.

Next, the "polluter pays principle" is an idea rooted in fairness. Those who pollute are held accountable for managing the resultant pollution. For example, industries that produce toxic waste are responsible for treating the waste before they can dispose it.

The third principle, to "pre-empt pollution and take early action", is derived from the fact that our understanding of the environmental impact of technology is far outpaced by its advancement, as the case of leaded petrol demonstrates. Thus, NEA watches closely for potential sources of pollution and takes pre-emptive action where possible. An example is Singapore's control of endosulfan, a pesticide harmful to health, in 2008. Three years later, this pesticide was added to the list of persistent organic pollutants governed by the Stockholm Convention, an international treaty to restrict organic environmental pollutants.

Armed with these three principles, everything else follows, from a legal framework to planning infrastructural developments.

LAWS FOR A BETTER ENVIRONMENT

Right up to the 1970s, laws and regulations in Singapore were still inadequate to protect the environment. For example, even though it was illegal to discharge matter that could be a nuisance or danger to health, no legal quantities were set. The provisions were also ineffective since it was debatable what constituted a nuisance or danger to health.

This began to change in the 1970s when several laws were introduced to improve public health and environmental protection. A key milestone was the Clean Air Act of 1972, which specified industrial emission limits, among other regulations. These laws were precise in their requirements and allowed better control over pollutive activities. Together with the establishment of ENV in 1972, dramatic improvement in Singapore's environment was accomplished in a matter of decades.

"The role of law in Singapore cannot be divorced from the administrative backbone to implement it. The fact that we don't have corruption on a systemic level means that companies can't just pollute and get away with it. Pollution control is partly technological. We have monitoring everywhere. In major factories, we literally can monitor what is coming out of the smokestack the moment it is happening. And I'm glad that we have moved beyond law as enforcement to co-enforcement. Now you can see a lot of major companies tracking their own pollutants. The environmental fact, the law as lived, is clearly strong."

— Professor Simon Tay, Chairman, NEA (2002 - 2008)



Over time, Singapore's environmental laws have undergone several rounds of revisions and consolidation to meet changing needs. Today, NEA administers a number of acts with their accompanying regulations, such as the comprehensive Environmental Protection and Management Act.

Under these laws, the private sector is also required to do its part in controlling pollution. For instance, any development with the potential to cause serious pollution has to conduct a Pollution Control Study. The study identifies sources of pollutants and recommends measures to prevent or reduce the impact of pollution. Industries also have to implement their own monitoring programme and ensure timely response to any incidents.

To make it easier for industry stakeholders to understand Singapore's environmental requirements, a guidebook, called the Code of Practice on Pollution Control (COPPC), was published in 1994. The code, which has been updated several times, lays down the requirements for every type of development, from industrial estates to business parks. It also shares good design practices with architects, engineers, industrialists and consultants.

A CASE FOR FLEXIBILITY

Did you know that for trade effluent to be discharged into sewers, its pH value must be between 6 to 9 and its temperature cannot exceed 45°C? NEA also sets the limits of substances in trade effluent such as sulphates, chlorides and heavy metals. These are just some of the requirements listed in the Code of Practice on Pollution Control to ensure that trade effluent does not cause undue pollution in the waterways. The regulations are necessarily demanding to make sure that our environment is not compromised by industrial activities.

The code does not specify how trade effluent, emissions and such are controlled. Instead, it lists the desired outcomes of pollution control. This approach was taken after consultations with industry stakeholders between 2000 and 2004. Businesses can decide on the best and most cost-efficient methods to control the pollution they create, and exercise creativity in coming up with systems of environmental management.



Making A Difference At The Onset

Have you ever wondered why your home is not placed right next to a noisy factory? Or why, if you are enjoying a meal at a hawker centre, the smells of industries are not part and parcel of your experience? Much of our daily lives go unaffected by industrial activities, thanks to intensive planning of land use by the government, a sound regime of inspection and enforcement, and NEA's close collaboration with the industry.

PLOTTING MATTERS

Land-use planning was adopted in earnest in 1971, when the first Concept Plan was launched. The plan mapped out broad strategies and long-term use of land for a period of 40 to 50 years, and has been updated every decade since. Details for development for the next 10 years, such as specific zones for housing or industrial development, are captured in Singapore's Master Plan.

The drafting and reviews of these plans are driven by the Urban Redevelopment Authority (URA), with inputs from various government agencies including NEA. This inter-agency collaboration allows for optimal use of land resources, ensuring that the environment remains clean, liveable and sustainable even as the economy develops.

Walk around the housing towns in Singapore and you would observe that within the estates, factories that may cause pollution are hardly to be found. Light industries are usually sited some distance away, and general and special industries operate only in purpose-built parks. This deliberate arrangement establishes distance buffers between industrial and residential zones to reduce the potential impact of pollution and nuisance on Singaporeans. Planning is carried out even for every plot within the zones. For instance, care is taken to locate compatible industries near each other. Industries sensitive to cross-contamination, such as food factories, will not be situated near general or special industries.

There are other factors taken into consideration too. For instance, facilities sensitive to noise, such as schools and hospitals, are located further away from expressways. In allocating land use around airports and airbases, the Noise Exposure Forecast (NEF), a measurement of exposure to aircraft noise, is used. Areas experiencing NEF *35* and above are used for industrial developments while residential development is allowed only in areas experiencing NEF below *35*.

Protecting our water catchment areas is particularly critical as water is a scarce resource here. Development in protected catchment areas is strictly controlled. In unprotected water catchment areas, only clean and light industries that do not use large amounts of hazardous chemicals are allowed, so as to minimise the risk of pollution to our water resources. Pollutive industries are located strictly outside catchment areas.

By planning far in advance, a balance is achieved between the demands of development and a high quality of environment.



"For every land use, we have to assess its suitability in terms of its potential environmental impact and dis-amenities that it can pose to the surroundings. Through the assessment, we will then state our environmental requirements to the developers to ensure that their activities can co-exist with the other activities in the area. The challenges that we face include meeting the expectations of industrial developers and the public, and to balance economic growth with our environmental objectives. This will be a greater challenge in the future, because of our limited land area coupled with continuous growth in population and industry. What we are doing now, results may not be imminent. Time will tell if we have done a good job. We often work behind the scenes, and at times we may only be perceived as having increased the operating cost of the industry. Nevertheless, if we can continue to enjoy clean air, clean land, clean water at the end of the day, I will call it an achievement!"

— Chow Chee Kiong, Senior Engineer, Central Building Plan Department, NEA

"Factories operate in different ways, and inspectors need to have a good understanding of the various industrial processes and the pollution control systems. The inspector must understand the factory, the production lines, and if the equipment fails, why it fails and what may go wrong. The inspector must also know the ground, the factory and its surroundings. For example, if a factory is near the water catchment area, the inspector must know how it will impact the water. Every day, every inspector carries out four to six inspections. When a factory is causing pollution, our inspectors also have to become witnesses for prosecution. They play a very key role in making sure that there is compliance."
— Jothieswaran Poobalasingam, Chief Engineer, Pollution Control Department, NEA

CONTROL, RIGHT AT THE SOURCE

Controlling pollution right at the source is a simple idea, but a complex act to follow through.

Before an industry is set up here, NEA first assesses if it is suitable for Singapore. Industries highly detrimental to the environment, and those generating significant amounts of toxic industrial waste, are not permitted even if they are able to add significant economic benefits to Singapore. Some examples would be paper mills and ore smelting plants. Companies bringing in obsolete, pollutive technologies are not welcomed either.

When the Economic Development Board (EDB) brings in potential investors for industrial projects, NEA informs these investors of Singapore's pollution control requirements. By laying out all the regulations in advance, NEA ensures that investors can make an informed choice on the suitability of Singapore for their projects.

As soon as an investor submits a proposal to build an industrial facility in Singapore, NEA will determine if the land parcel is the right fit, taking into account the needs of nearby users. Once the proposal is approved, a process of licensing and inspection will kick into motion. NEA officers will visit the site, ensuring that from design to completion, measures are in place to control pollution. In an average year, NEA receives around 100,000 related applications, from issuance of licences to temporary occupation permits.

A RIGOROUS REGIME

Without a rigorous regime to administer our laws and regulations, there is no way for environmental protection to be truly effective.

First, NEA uses licensing to track and regulate the use of materials which may impact the environment. Licences for factories or usage of hazardous substances, for example, are only issued when applicants show that they are adhering to legal requirements, such as having the necessary pollution control equipment and containment facilities in the premises.

Next, the agency's factory inspectors carry out regular checks on industrial premises to ensure that there is compliance with regulations on the ground. If the premises are found to be flouting requirements, enforcement measures will be taken. These measures range from the issuance of warning letters to prosecution in court, depending on the severity of the violations.

Finally, NEA practises constant vigilance to keep pollution in check. The idea is simple – watch the health of the environment closely and pollution can be nipped in the bud before it escalates. The constant monitoring of our air and water not only provides an up-to-date report card of the environment, it also allows NEA to detect and respond to pollution incidents quickly. More importantly, it helps NEA to track the effectiveness of its programmes and policies in protecting and improving our environment.

"In the early days, you could find sawmills at Lavender Street! Can you imagine, a sawmill at Lavender Street? And people were living next to it. That was how it was, because there was no need for people to get permission to set up a factory. But slowly, this changed. After the Anti-Pollution Unit was formed, any industries that came in, they had to get approval for the piece of land and we would see whether or not it has any impact on the environment. Also at the same time, we would see if this industry is suitable for Singapore. Because Singapore is so small, we cannot have industries that generate a lot of waste. And certain industries use a lot of water, so they are also out, because Singapore is very short of water. And if an industry generates waste that cannot be treated in Singapore, let's say radioactive waste, or some waste so toxic that we don't have the facilities to treat them, we would not permit the industries to set up here. So firstly, whether the industry is suitable. And if the industry is suitable, whether the siting is suitable. For example, we cannot allow sawmills to be in residential areas. Then steel mills cannot be sited next to food factories. So within the factories, we must also plan, otherwise there may be cross-contamination. And we make sure that industries sited close to each other are compatible. After we look at the site, we look at processes. We don't want a company to bring in pollutive, outdated or obsolete processes and technology to Singapore and generate a lot of pollution. So we will make sure that they will bring in the right kind of technology and pollution control equipment."

— Joseph Hui, Deputy CEO, NEA





A STORY OF PLANNING

Think about it. The tiny island state of Singapore, without oil resources of its own, is one of the top petrochemical hubs in the world.

What is even more interesting is how the quality of Singapore's environment remains one of the best in the region, even with over 100 companies including leading petrochemicals companies that are heavily pollutive in nature. Talk about having your cake and eating it too.

The key to the puzzle lies in Jurong Island, located just a few kilometres off the shore of mainland Singapore. The result of land reclamation carried out in the 1990s to form a land mass out of seven smaller islands, Jurong Island is located offshore to minimise the degradation of the quality of environment and impact of potential accidents on mainland Singapore. Reclamation works began under Jurong Town Corporation (present-day JTC Corporation) in 1995. Working closely with ENV (present-day MEWR) and other government agencies, JTC took care to incorporate facilities for pollution control in Jurong Island's infrastructure.

Today, Jurong Island is the premier hub for the chemical industry in the region, with one of the largest oil refining and fuel storage terminals in the world. Even more impressively, its air and water quality, monitored by NEA, is within international standards and consistently comparable to that in mainland Singapore.

The importance placed on planning for the environment is all the more pronounced in the latest initiative rolled out by the government – Jurong Island Version 2.0. More petrochemical companies are expected to set up here, which means that several environmental challenges are posed. New plants and chemical pipelines will have to be located closer to existing facilities. As such, chances of a "knock-on" effect, whereby an explosion from a plant may cause secondary explosions or accidents in neighbouring plants, are higher. The storage, transportation and use of hazardous chemicals will be significantly increased. There will also be more sources of pollution. The consumption of resources such as electricity and water, and generation of waste will also go up.

Driven by EDB, Jurong Island Version 2.0 aims to address these new challenges as well as improve existing environmental infrastructure. Multiple agencies, including NEA, are involved in the project. NEA plays a critical role in the realisation of Jurong Island Version 2.0 by providing technical expertise and advice. It works with other agencies to develop detailed guidelines and safety procedures to address the "knock-on" effect in potential accidents. More stringent control of air emissions from the industries is looked into to make sure that air quality in mainland Singapore and Jurong Island will not deteriorate. Conserving resources being a priority, NEA advises the agencies involved on reducing and recycling waste and maximising use of waste energy, such as the possible conversion of waste heat into electricity. NEA has also commissioned consultancy studies to study waste management, treatment and utilisation in Jurong Island.

By ensuring that such measures are integrated in the planning and development stage, a more competitive, self-sustainable Jurong Island Version 2.0, one of the few petrochemical hubs in the world where the quality of environment remains safe and high, is set to be realised.

INNOVATING ON ONE HAND, ROPING IN THE INDUSTRY ON THE OTHER

With the number of industrial premises, vehicles and developments in Singapore, making sure that each of these complies with environmental regulations is a great challenge. To address this, NEA constantly explores new technology, employing innovation to improve and enhance operations. This approach can be seen in all aspects of NEA's work in environmental protection, from remote monitoring of pollutive emissions to investing in informaton technology systems to manage operations.

Another key strategy to control pollution right at the source lies in roping in the industry to do its part. The industry is closer to the ground and can better assess whether certain solutions are viable. The government can also free up resources for other uses if the private sector takes more initiative at its end. Most importantly, this will foster a culture in the industry whereby environmental management becomes an integral part of business concerns.

One way in which NEA engages the industry is through regular consultations. During such sessions, both parties share insights and put forward their points of view. With the feedback, NEA can gauge if its policies and regulations are feasible, and implements changes where necessary. NEA also actively encourages industries to take on more self-policing and strive to reach higher than stipulated standards. For instance, the agency supports the Responsible Care Programme launched by the Singapore Chemical Industry Council to promote good management and handling of chemicals.



"ExxonMobil enjoys a collaborative working relationship with the National Environment Agency through the years. The agency makes an effort to engage with industry players and to understand the complexity and specific requirements of the refining and petrochemical industry, as well as its associated needs."

— Derk Jan Hartgerink, Manufacturing Director, Singapore Chemical Plant, ExxonMobil Chemical Asia Pacific

The Drive For Clean Air

Breathe in, deep. And enjoy just how fresh the air is. Despite having to share a small island with a large number of factories and vehicles, taking a deep breath is rarely a health hazard for Singapore's residents. NEA plays no small role in bringing this about. It tracks industrial emissions, enforces against polluters, implements higher benchmarks for air quality and monitors the health of our ambient air round the clock.

"NO" TO SMOKE ON THE ROADS

Vehicles, factories and power plants – these are the key sources of air pollution anywhere in the world. What makes vehicular emissions all the more dangerous is the fact that pollutants such as carbon monoxide, nitrogen oxide and particulate matter are released at ground level, mixed in with the air we breathe.

Restrictions on emissions have become increasingly stringent as the years passed. For instance, emission standards for motorcycles and scooters were further upgraded in 2003. And in 2006, the Euro IV emission standards, which demand lower sulphur content in diesel, were imposed for all new diesel-powered vehicles.

Fuel quality is also tackled by the agency. Take sulphur content. Sulphur is naturally found in crude oil, which is refined into diesel fuel. The substance is largely responsible for the release of particulate matter and sulphur dioxide, both of which are pollutants. In 2005, NEA reduced the permissible sulphur content in diesel to 0.005% by weight to pave the way for ultra-low sulphur diesel.

Imposing such requirements involves a lot of work. For the implementation of Euro IV standards, NEA had to consider issues such as the availability of Euro IV compliant vehicles, and social and economic costs. Generally, manufacturers need about two years to roll out new designs of vehicles equipped with more efficient engines and pollution control devices. Another two years are needed for local car dealers to clear existing stock and bring these newly-improved vehicles to Singapore. Altogether, it would take a few years before a more stringent standard can be effectively implemented.

NEA also enforces penalties upon those who drive and who own smoky vehicles. In the past, this would be manually performed by officers armed with smoke meters. Nowadays, officers equipped with video cameras stake out the roads. They capture smoky vehicles on video, scanning at least 5,000 vehicles a day. Drive past the checkpoints at Woodlands and Tuas, and you may very well see officers on the lookout as well. Malaysia-registered vehicles are also subject to the regulations and those emitting smoke measuring more than 50 Hartridge Smoke Units (0 HSU is clear while 100 HSU is opaque) are fined. Those emitting smoke more than 70 HSU are not allowed to enter Singapore.

Drivers who leave their engines idling are taken to task, except under specific conditions such as taxis and buses waiting in a moving queue. This is to minimise the emission of exhaust gases, odour, noises and fuel waste caused by idle engines.

Education also plays a key role in NEA's efforts to combat vehicular emissions. The agency works with large vehicle fleet owners such as taxi and bus companies and the National Association of Travel Agents Singapore to educate drivers to turn off their engines while having meal breaks so as to prevent idling engines. There is also more public feedback on idling engines as Singaporeans become increasingly aware of the harmful effects of pollution caused by inconsiderate motorists.

As a result of NEA's efforts, the number of smoky vehicles on the road has dipped considerably to about 0.6% of the vehicle population in 2011 from about 5.8% in 1990.



PASS THE TEST FIRST!

No person shall use or cause or permit to be used on the road any motor vehicle from which any smoke or visible vapour is emitted.
— Regulation 19 (1), Environmental Protection and Management Act

If it is smoky, it is not going anywhere. Every vehicle in use is required to go for a smoke test during its mandatory periodic inspection at an authorised inspection centre, which checks for the amount of carbon monoxide and smoke released. Depending on the type of vehicle and year of registration, permissible emission limits vary. Any vehicle that fails the periodic inspection is not allowed to renew its vehicle road tax until it passes.

Today, all authorised inspection centres adopt a procedure known as Chassis Dynamometer Smoke Test (CDST) to measure the smoke emissions of diesel vehicles. Piloted in 2000 and made mandatory since 2007, the test simulates actual driving conditions, making the test more accurate than the previous procedures used. This enables NEA to be more effective in its inspection, and further reduces the number of smoky vehicles on the road.

A 24/7 Watch

Fifteen ambient air-monitoring stations around the island, sensors installed in the chimneys of factories, video cameras trained on smoke stacks in industrial parks – NEA keeps an eye on signs of air pollution 24/7.

This continuous monitoring achieves some important goals. It allows NEA to pick up on any deterioration of air quality straightaway and respond immediately. There is ready evidence for these acts of non-compliance when it comes to enforcement against culprits. And NEA can know for certain if its policies and programmes to ensure good air quality are effective.

Up to the mid-2000s, monitoring industrial air emissions was still a time-consuming task. NEA's engineers had to scale structures near the towering smoke stacks to extract air samples. The samples would be put through a series of laboratory tests to check for impurities. Such a system meant that NEA could only assess the quality of air emissions at a particular time, whereas the contents varied through the day.

As technology advanced, better solutions were invented, such as sensors which could analyse air pollutants continuously. Pollutive industries were required to install such devices to track their own emissions. To monitor emissions from these industries, NEA hit upon the idea of linking these sensors up to a central data network. This would give NEA specific, real-time information on industrial emissions. The system piloted in 2007 and officially kicked off in 2010. Major emitters of air pollutants such as power plants are required to install the monitors.

NEA also carries out round-the-clock visual surveillance on the smoke stacks of industries. Video cameras scan key industrial areas for visible smoke. With this, NEA can address public feedback on smoke emissions immediately, as the agency can instantly identify the offending smoke stack using real-time video. Officers can be deployed quickly to ascertain the ambient air quality in the area as well.

Equipped with such an extensive surveillance system, NEA's efforts in ensuring good air quality for Singaporeans are now much more effective and efficient.

READING THE AIR

"Monitoring allows us to find out where we stand in comparison to international benchmarks for the environment. We have formulated air pollution control strategies and programmes to address air pollution sources. But even as we put in place stricter regulations, the sheer numbers of vehicles, our demands for energy, urbanisation and industrialisation pose real challenges. And as international standards become higher, it becomes even harder to reach the new targets. At the same time, pollution control measures come at a cost. So we have to strike a balance. All of us have a part to play."

– Indrani C Rajaram, Chief Scientific Officer, Environmental Monitoring and Assessment Unit, NEA

On the surface, it sounds really simple. Sophisticated hi-tech analysers are fitted in 15 ambient air monitoring stations island-wide. They detect the levels of six main pollutants in ambient air – carbon monoxide, nitrogen dioxide, ozone, particulate matter 10, particulate matter 2.5 and sulphur dioxide. The data, stored in a central network, allows Internet-based access for NEA officers.

Certainly, compared to the 1970s and 1980s, tracking ambient air quality is now more efficient. Formerly, officers had to manually collect samples and conduct laboratory tests to determine the air quality. Automated continuous monitoring of ambient air kicked off in 1994 with the installation of a telemetric system. Upgrades were carried out in 2005 to enhance the system. Today, air quality is tracked at ambient and roadside stations. There is even a mobile air monitoring station to aid on-site investigations.

But even with such a sophisticated system, monitoring the air still requires a strong team of scientific and technical experts. Indrani C Rajaram,







Chief Scientific Officer, leads NEA's team in monitoring and assessing the performance of Singapore's air and water quality. Says Indrani, "The instruments we have are highly specialised equipment that have to be calibrated and checked regularly, and the data they produce must be validated constantly. This is to make sure that what the instruments are giving us are not aberrations, but accurate measurements of the pollutants in the air. There is quite a lot of work to be done."

Indrani's team is behind the daily Pollutant Standards Index (PSI) reading we hear every day. The PSI uses a scale ranging from 0 to 500 to measure air quality. At readings of 50 and below, the quality of air is rated as good, while any reading above 100 indicates that the ambient air is unhealthy. Armed with such information, members of the public can decide if they should undertake certain activities such as outdoor sports. This service becomes even more critical during events such as the periodic haze.

"We had to deploy people round the clock to report the PSI reading every hour during the haze," recalls Indrani. "The PSI reading is based on the worst concentrations of air pollutants, because the reading is for members of the public to take the necessary precautions to protect themselves from the harmful effects of the haze. So we take the conservative approach. We compute the PSI reading based on the highest concentrations of the air pollutants using data from all the ambient stations, even if there are pockets of Singapore where the concentrations may be lower."

As NEA is the national authority on the quality of air, the team also plays a critical role in assuaging public concerns when the haze comes around. "Coping with the sheer number of queries was a big challenge!" says Indrani, remembering the haze in 2006. "The public was very concerned. We had callers wanting to know what NEA was doing about the quality of air, people who had doubts about the PSI readings and such. Many of them wanted to know if they could carry on with their normal routines even with the haze. So in between working round the clock in shifts to report the PSI readings and making sure that our equipment was performing all the time, our team had to answer these queries and allay their concerns."

What is even more challenging is helping to ensure that Singaporeans continue to enjoy a high quality of air even as the country develops. Indrani and her colleagues support NEA's drive for better air quality by contributing to new, higher benchmarks for Singapore's environmental performance and constantly improving their monitoring competencies. The unit is looking at launching more powerful systems to monitor the environment and even real-time sensors for our coastal water quality. To Indrani, who has been in this line since 1978, these challenges are exciting opportunities to find better solutions. She shares, "There has never been a dull moment. Work here has been so exciting! We know that whatever little we can do is going to help. When new standards for air or water quality are announced, we know that we are the guys behind coming up with these standards for Singapore. That's really rewarding."

"October 2006, in the middle of the fasting month. The haze was really very bad. There was a breaking of fast I attended and on the way back, I drove down River Valley Road. I couldn't see in the night! So the next day, I said, we need to do something about it. So we called for a regional meeting."

– Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011)

ALL THE WAY TO JAMBI

It was one of the worst periods of haze Singapore had ever experienced. In October 2006, the city was shrouded in grey smoke. The smell of burning permeated the air. On 7 October, the PSI reading shot up to a high 128, well in the unhealthy range. NEA's advisory for the day read, "The smoke haze situation has deteriorated overnight ... The latest satellite pictures showed that there were 506 hotspots and thick smoke haze in Sumatra ... persons with existing heart or respiratory ailments to reduce physical exertion and outdoor activities. The general population should reduce vigorous outdoor activity."

This was not the first time Singapore had experienced the haze. The last bad onslaughts were in 1997 and 1998. Periodically, when the southwest monsoon comes around between June and September, it brings the dry season along. In Borneo and Sumatra, peat fires, as well as those started by farmers using the "slash and burn" method to clear the land, spread fast and more intensely during this period. The situation is further exacerbated when it coincides with the El Niño phenomenon, which usually leads to an extended dry spell. The fumes from these fires are carried by prevailing winds to Singapore, Malaysia and parts of Thailand.

In 2006, the situation was critical. There were significant health risks posed to Singaporeans. The quality of life and economy suffered too. Given the seriousness, Prime Minister Lee Hsien Loong wrote to President Susilo Bambang Yudhoyono on 10 October to express concern. Singapore also called for an urgent regional meeting of environment ministers from the countries affected by the haze.

The meeting took place on 13 October in Pekanbaru, Indonesia. Following the meeting, ministers from Singapore, Brunei, Indonesia, Malaysia and Thailand formed a steering committee to oversee initiatives to address the haze. They also decided on meeting once every three months to oversee a plan of action.

Senior officials from ASEAN member countries, international and regional non-governmental organisations (NGOs) and inter-governmental organisations came together in November to brainstorm and develop a comprehensive plan. In the plan, 35 fire-prone areas in eight provinces of Indonesia were identified. Next, Indonesia called for ASEAN member countries and international organisations to aid these provinces in building up capacities to prevent and fight land and forest fires. A technical working group, with members from the affected countries, was formed to support such initiatives.

For Singapore, it was clear that if the initiatives were to be effective, they would have to be implemented at ground zero, right where the fires





Left: Initiatives, such as this ministerial steering committee workshop in 2011, are regularly held to address the haze issue.
Right: An ambient air monitoring station installed by NEA in Jambi, Indonesia.

took place. Hence, the government decided to respond to Indonesia's call and offered its help in Muaro Jambi Regency. The regency was selected by the Jambi Provincial Government as it is fire-prone, and its proximity to the city area means that many residents are adversely affected when the haze happens.

Together with the Jambi Provincial Government and Indonesia's State Ministry of Environment, Singapore developed a master plan to deal with land and forest fires. A team comprising officials from MEWR and NEA was sent to Jambi in January 2007 to understand the ground better. After much research, the master plan was finally drafted and finalised in March 2007.

The plan outlined strategies for sustainable land-clearing practices, fire prevention and suppression. Specific programmes were also designed to realise the plan's objectives. While the Jambi Provincial Government was responsible for implementing the programmes, it also received assistance from Singapore and other ASEAN countries as and when it made such requests. At the invitation of the provincial government, Singapore signed a Letter of Intent on 7 November 2007 to help execute a number of programmes, setting aside $1 million to do so. The key agency representing Singapore in these efforts was NEA.



NEA officers with locals in a village in Muaro Jambi Regency.

GROUND UP EFFORTS

"It impressed upon me very strongly that the haze is not just an environmental problem, and it cannot be addressed from the environmental aspect alone."

– Rohaya Saharom, Deputy Director, Pollution Control Department, NEA

Rohaya Saharom was one of the engineers sent by NEA to Jambi to help tackle the haze issue. A member of the fact-finding mission in 2007, her technical expertise and fluency in the Malay language were critical assets on the ground. She recalls, "Once we were there, we could get a sense of the magnitude of the problem. The district is very big, and deployment of resources is a challenge. Some of the fires took place deep in the forests and gaining access to them was difficult. There was also the question of where to draw the water for fire-fighting from. Besides, the issues were complex, affecting livelihoods. Clearing land using fires was also attributed to certain sectors having lack of access to land-clearing equipment such as bulldozers and tractors."

Together with her team, she made more than 10 trips to the regency between 2007 and 2009 to help launch the programmes chosen for joint implementation. These programmes were carefully planned to develop specific capacities in Jambi officials, increase their situational awareness of fires and hotspots, improve their system of fire prevention and suppression and educate local farmers on more sustainable agricultural practices.

"We organised workshops for the locals, followed up on projects and worked with the authorities and NGOs," says Rohaya. "Forest conservation, peatland management and forest fire prevention and suppression – these are subject areas outside our usual scope of work, an unfamiliar territory for us. So we engaged experts and consultants who were field practitioners in these areas to assist in activities to build capacity and raise awareness on fire prevention and mitigation, and peatland management. The villagers were very enthusiastic. Many were concerned about the state of their land, and the floods they were experiencing. The experts shared how the improper use of land could lead to such consequences. We had a consultant to review the fire-fighting capacities of the locals and present recommendations on bridging the gaps. To me, these efforts were meaningful because we were working from the ground up."

NEA also trained the Jambi officials in reading and interpreting satellite imagery and hotspot information, ambient air quality monitoring and using Geographical Information System to support fire and haze monitoring. Three air and weather monitoring stations were set up so that residents could know the state of their air quality. Land-use maps were developed to help Jambi officials identify and monitor areas affected by fires. "With the ambient air monitoring stations, the locals are able to know the state of the air quality and take the appropriate precautions to safeguard their health during a haze," shares Rohaya.

With strong support from the Indonesian authorities, the programmes were successfully implemented by 2011. However, the regency is only one of the many fire-prone provinces in Indonesia and the scale of the issue remains considerable. Nonetheless, the joint efforts have made a difference in one regency, and in time to come, they may be replicated in other fire-prone provinces in Indonesia.

A Resource To Protect

Water is the essence of life, and a scarce resource in Singapore. This makes NEA's work all the more crucial as it monitors the quality of raw water streams in our catchment areas, ensuring that it is safe for treatment into drinking water. Our coastal waters, especially those around recreational beaches, do not escape the agency's watch either.

BE IT INLAND OR COASTAL WATERS

Every Sunday, it is either kayaking at Sembawang Park Beach or dragon-boating at Kallang River for Eddie Foo and his friends. To Eddie, contaminated waters are the last thing on his mind when he is immersed in these activities. "I grew up in an era when we sometimes swallowed mud and sand while playing! So this is just part and parcel of life; if you want to go onto water, you will sometimes swallow water and that beats mud any day," says Eddie. "I'm not afraid of contamination. It's not that I don't care. If I see pollutants or rubbish in the water, I will try to steer clear of it. But if I take a spill, and I have no choice, I just take it."

But thanks to NEA, if indeed the recreational beaches, inland and coastal waters in Singapore are contaminated, beach goers and water sports enthusiasts will be warned against entering the waters.

Few are aware of this. NEA commissions the weekly sampling of water quality around six recreational beaches (Sentosa, Seletar Island, Sembawang Park, Changi, East Coast Park and Pasir Ris Park) and inland waterways. There are also 9 sampling points set up in the Straits of Johor and 10 in the Straits of Singapore.

Samples taken from coastal areas are tested for the pathogen, *Enterococcus*, and fresh water samples are tested for both *Enterococcus* and chlorophyll-a. If the amount of *Enterococcus* exceeds 200 counts per 100 ml of water in more than 5% of the samples over a one-year period, the site will be declared unsafe for activities such as swimming, where users have primary contact with the water. Chlorophyll-a has to be below 50 µg per litre of water.

This is done for a simple reason. *Enterococcus*, found in the faeces of humans and warm-blooded animals, causes gastrointestinal and respiratory illnesses such as diarrhoea and throat irritation. Chlorophyll-a, on the other hand, is an indicator of algal blooms in a fresh water body. The blue-green alga produces toxins harmful to humans and animals. People may fall ill if water contaminated with these substances gets into their mouths, noses, eyes or open wounds. When they take part in water activities, chances of this happening are higher.

Besides testing for biological contaminants, NEA also analyses the water samples using physical and chemical parameters such as the level of dissolved oxygen and amount of suspended solids. This is done to affirm that our waterways and recreational beaches are healthy and safe.

Should any of the water bodies under its purview fail the guidelines, NEA will investigate the causes of pollution, and take necessary action to remedy the situation.

Using these parameters, NEA has found that the quality of water in Singapore's beaches and inland water bodies has not deteriorated over the years. As of 2011, the Ministry of Health (MOH) has not received any reports on disease outbreaks due to poor recreational water quality. The efforts of NEA and other agencies in monitoring and maintaining our water quality have allowed Singaporeans to use the beaches and waterways with peace of mind.



Guidelines for measuring the quality of water for recreational use were implemented in August 2008, based on a World Health Organization matrix. The decision to do so was taken after an extensive study by an inter-agency committee comprising NEA, PUB, Maritime and Port Authority of Singapore and the Ministry of Health, as well as advisors from the National University of Singapore and Nanyang Technological University.

SAFE TO SPLASH!

If Pasir Ris is one of your favourite hangouts, you would remember the years when signs advising beach goers against swimming in the sea lined the beach. Those signs were put up on 1 August 2008 by NEA after it launched more stringent guidelines for water quality based on WHO standards. Under the new guidelines, the waters at Pasir Ris Beach were no longer deemed healthy for activities where there was contact with water. This was because only 93% of the water samples taken over three years showed readings of less than 200 counts of *Enterococcus* per 100 ml.

Besides warning the public against swimming, NEA also advised beach goers to take precautions, such as avoiding water sports if they had open sores, skin infections or were unwell, not to touch their mouths and eyes with hands soiled with sand, and to wash their hands thoroughly before handling food.

In the same year, NEA also commissioned a study to investigate the origin of pollution affecting our recreational beaches. The results showed that our beaches were mainly affected by local sources. In the case of Pasir Ris Beach, the key culprits were sewage treatment plants and leakages from the sewer network. Besides, water currents in the concave part of the Pasir Ris Beach were weak and pollutants were not as quickly dispersed and diluted as in other beaches.

The aging sewers in the Pasir Ris Beach area were repaired under PUB's sewer rehabilitation programme. PUB also launched a plan to phase out the sewage treatment plants.

In 2010, the water quality review showed that the amount of *Enterococcus* at Pasir Ris Beach was again too high. Seven per cent of the samples showed *Enterococcus* counts of higher than 200. Though the measures had taken effect some time after implementation, NEA decided to take extra precaution and continued monitoring the situation to ensure that the water quality was stable. The beach was finally opened to the public in January 2012.

"I love the sea. I like how immense it is, how elemental and primal. In the case of Singapore, there is nothing quite like being able to run into the water after a day at work. That is what the sea and coast mean to me – it is a taste of freedom and getting away from it the pressures of the rat race and other mundane troubles for a time. I have swallowed water while swimming and have wondered whether the water is safe. Given the amount of shipping traffic Singapore accommodates, it is very important to know if our waters are safe to swim in. I think such sampling done by NEA is absolutely vital!"

– Alexandra May Serrenti, a frequent user of East Coast Park Beach



Samples are collected at a depth of 1 m in the sea. Officers have to take proper precautions to prevent the water from entering their mouths, ears or nose.

Keep The Noise Down!

The sounds of people conversing, vehicles on the roads, television and radio sets, hawkers busy at work in coffee shops and hawker centres – such is the usual medley of noise we have to live with every day. NEA's work is crucial in striking the balance between urban activities and a high quality of living environment. Materials used to pave roads are noise-absorptive to some extent. Industrial buildings, multi-storey car parks and parks are located between major roads and residential areas to act as noise shields. New vehicles must meet noise standards before they can be registered and vehicles-in-use are tested for the noise they emit. These measures help to off-set the amount of noise in our urban environment.

SHHH ... IT'S SUNDAY

A construction site near your home? Thanks to a regulation set by NEA in 2011, you no longer have to put up with the maddening, mechanical noise of piling and construction work on Sundays and public holidays. In fact, by 10 pm the night before, peace would have set in.

Construction sites are never really pleasant. Besides noise, they generate dust, debris and odours. But these are a must if Singapore is to develop and progress. To ease the impact of construction noise on the public, NEA has put in place several measures.

On 1 October 2007, NEA implemented new regulations to lower the noise limits permitted from construction sites for night-time, Sundays and public holidays. The maximum fine for construction noise violations was raised from $20,000 to $40,000. Contractors who repeatedly violate noise limits would also be restricted in their operation hours.

Another round of review was held in 2010 and more stringent measures were imposed on sites located within 150 m of residential premises and noise-sensitive premises such as hospitals. For such sites which commenced work from 1 September 2010 onwards, construction activities were not allowed from 10 pm on Saturdays and eves of public holidays to 10 am on Sundays and public holidays.

The prohibition of work on Sundays and public holidays was further extended to new sites which began work on or after 1 September 2011. Since then, no construction work is allowed on these sites from 10 pm on every Saturday and public holiday till 7 am on the following Monday or day after a public holiday.



A handheld noise meter is used to gauge noise emitted at a construction site during inspections.

A REMOTE WAY OF SURVEILLANCE

Gone are the days when NEA's inspectors had to stand outside construction sites, a noise meter in hand to gauge if noise pollution regulations were being flouted.

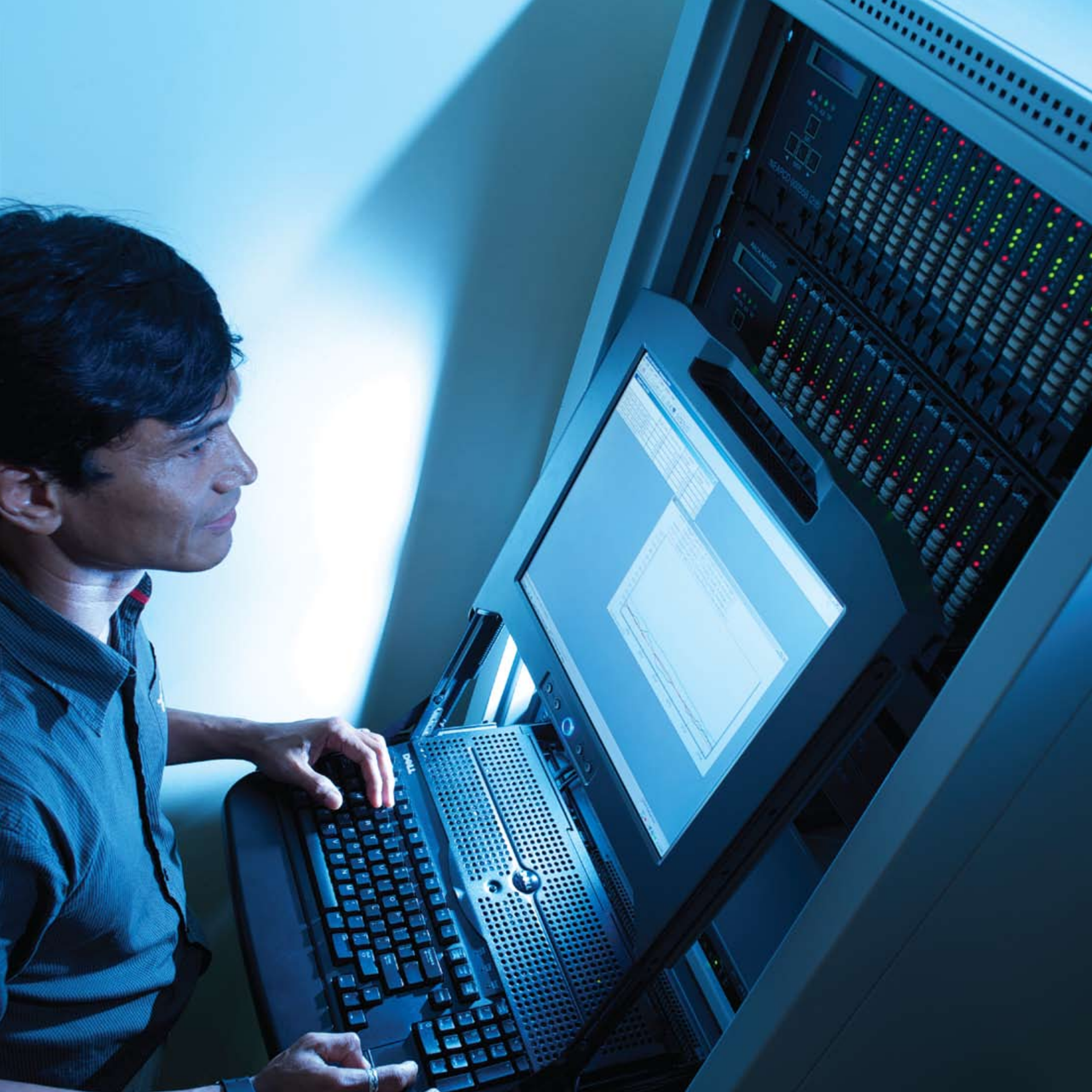
Inspectors can now keep a ear over construction sites all over Singapore, and know who's breaking the rules.

Since 2008, NEA has put in place a smarter, faster and better system. Solar-powered noise meters fixed to a residential building nearest the construction site pick up the noise emitted continuously. The real-time data is relayed via wireless communication to a central database. Both NEA and the construction companies can access the database online any time of the day to check on noise levels. When the levels exceed legal limits, the times at which the offences take place are highlighted in the system. Alerts will also be sent out to NEA and to the company.

The idea for such a system was hatched by NEA and the solution was developed by industry players. In the past, NEA would have to despatch an officer on-site to investigate every public complaint on construction noise. With such remote surveillance in place and more companies adopting the system, such work has been cut down dramatically. As of 2011, any construction project with a value of over $3 million is required to use the system.

"Remote monitoring is good both for NEA and for the contractor. Contractors can monitor and see for themselves the levels of noise being generated. In the past, using stand-alone meters, the data would only be downloaded at the end of the week. So through the week, they would not know whether the levels were high or low. On our end, once the complaints from the public come in, we can check the real-time data from the site in question, so we can update the complainants straightaway. Public complaints do not necessarily translate into violations. It may be higher expectations of the public. For us, it is important to establish for each complaint the noise level generated, whether it is exceeding the permissible limits. If it does, we will take the necessary enforcement action."

— Khairul Sani Samsudin, Senior Engineer, Pollution Control Department, NEA



Containing Hazards

They may be toxic, flammable, poisonous or highly corrosive. But in many ways, they are necessary. Many things in our daily life are linked to chemicals, from toothpaste to fuel and processed foods. Hazardous chemicals are also used on a daily basis in our chemical industry. With so many chemicals around, it is important that they are properly regulated to protect the public. NEA is in charge of the control and regulation of hazardous substances.

CHECKING HAZARDOUS MATERIALS

In such a densely-populated urbanised city, hazardous materials must be well-controlled to curtail the impact of potential accidents and abuse involving the substances. NEA regulates more than 500 chemicals designated as hazardous. One of the agency's key priorities is to ensure that residential and water catchment areas are protected from these substances.

NEA relies on a tight regime of licensing and inspection for regulation. First, any user who intends to import, export, store or use the chemicals must be licensed by the agency. The user must conduct an assessment to confirm that the chemicals do not pose health and safety hazards during its use, and provide mitigation measures during accidents. Those who handle the chemicals must also go through NEA's course on the management of hazardous substances and pass an examination before they are licensed. NEA also requires specific storage conditions and adequate pollution control measures. For instance, nitric acid must be stored separately from alkaline substances to prevent violent reactions. The requirements help to contain the impact of potential accidents within the premises so that they will not affect surrounding areas.

To make sure that these requirements are complied with, regular inspections are carried out. Public feedback and tip-off on lapses in factories will be investigated as well. With over 2,000 companies under its purview, NEA turns to risk-profiling to deploy its resources more efficiently. Companies are classified into one of three categories, depending on the amount and type of chemicals kept on-site. Category 1 companies pose the highest risk and are audited and inspected more frequently than the companies in the other two categories.

Companies which flout rules may be warned, given a show cause letter or even prosecuted if they are consistently recalcitrant. Should an accident occur, NEA will work alongside other government agencies such as Singapore Civil Defence Force (SCDF) and Ministry of Manpower in a coordinated response. NEA monitors the environment to assess the accident's impact. It also makes sure that the company cleans up the place to ensure the safety of its neighbours and workers.

Additionally, companies generating toxic industrial waste have to register with NEA and report the volume of such waste produced and where it is treated. The collection and treatment of toxic industrial waste can only be undertaken by NEA licensed toxic waste collectors.

Certain off-the-shelf products that contain chemicals exceeding stipulated limits come under scrutiny too. NEA conducts checks regularly. If products are found to contain controlled substances, or



if the levels exceed exemption limits, action will be taken against the importers.

Exceptional care is taken when it comes to transport of hazardous substances. Unlike fixed installations, where safety features are in place and workers are trained to handle hazardous substances, there are very few measures that can be undertaken to mitigate any incident that occurs during transportation.

To reduce the risk of an accident, companies have to seek NEA's approval to transport hazardous substances that exceed a stipulated quantity. Drivers must undergo a relevant course run by SCDF before they are permitted to transport hazardous chemicals. The routes and timing are restricted as well. Sundays, public holidays and peak traffic hours are to be avoided. This is so that if accidents do take place, their impact will be manageable.

In addition, tracking devices and remote-activated brakes are installed in the vehicles used for transporting highly toxic substances. This allows the location of the materials to be tracked at all times and prevents the vehicles from being used for sabotage.

"When we conduct inspections, we need to know which chemicals are stored at the companies, how they are used and whether the storage areas are secure. Proper documentation and pollution control equipment must be in place. We also need to educate businesses that it is essential to install pollution control measures. Sometimes, there are companies which do not understand why our requirements are so stringent. Because the chemicals they use may be cheap, but the pollution control measures are very expensive. For example, any company that stores more than 100 tonnes of hazardous substances must install an underground retention basin. The basin is meant to contain the water used for fire-fighting if fires do break out, so that the contaminated water does not pollute our land or water catchment areas. This basin may have to be as big as a swimming pool. So here they have to build a pool which must be kept empty and it does not seem to make sense. So we need to educate them, because we want to protect lives and the environment."

— Sulaiman Bin Abdul Rahman, Technical Support Officer, Pollution Control Department, NEA


TOXIC
6

BRIDGING THE SAFETY GAPS

It was just another Thursday morning. But if you happened to be one of the motorists using the Tuas Second Crossing on 28 July 2011, you would have seen industrial drums scattered on the crossing and emergency response personnel from SCDF and BOMBA Malaysia containing a mock chemical spill in earnest. Dummies were rescued from what appeared to be a multiple collision of an isotanker, a lorry, a passenger car and a motorcycle.

What took place was a joint field exercise between Singapore and Malaysia, during which both countries ran through emergency response operations in a simulated accident involving hazardous chemicals on the crossing.

Such exercises are critical as the transport of all hazardous chemicals between Singapore and Malaysia takes place on the Tuas Second Crossing. Every year, about 111,000 tonnes of hazardous chemicals, such as sulphuric acid, sodium hydroxide, phenol and formaldehyde, are transported between the two countries.

If you take a look at the areas surrounding the Causeway at Woodlands, it is clear why the transport of hazardous substances is confined to the Tuas Second Crossing. Woodlands, Bukit Timah and Johor Bahru are all congested and densely-populated zones, while Tuas and the western part of Johor are relatively less populated. By using the Tuas Second Crossing, the potential impact of accidents on residential areas is reduced.

The emergency response plan is the result of planning and discussions among the Malaysia-Singapore Joint Committee on the Environment, the Department of Environment, Johor and NEA. A procedure is set up to alert both countries as soon as possible when an accident takes place so that action can be taken early on. To make sure that agencies on both sides can coordinate their operations and get the situation swiftly under control, a detailed response procedure is laid out as well. Joint field exercises are held regularly so that all the agencies involved are familiar with the procedures, with Singapore and Malaysia taking turns as organisers.

Between 2000 and 2011, seven exercises were conducted. A number of agencies from Singapore and Malaysia are involved in every exercise, including emergency response forces such as SCDF and BOMBA Malaysia. Chemical companies were also invited to take part by providing emergency response personnel, vehicles, drums and other props.

During every exercise, different scenarios involving vehicles, motorists and chemicals are enacted to test the responses of both countries in various aspects. For example, in the 2011 exercise, the accident was simulated 500 m away from the Singapore Tuas Checkpoint. The site was under Malaysian territory even though it was nearer to Singapore, and the Singaporean team arrived at the scene before the Malaysian forces. The exercise thus put to test Singapore's ability to effectively hand over command of the operations to Malaysia in a time-critical scenario. SCDF and BOMBA Malaysia also had to race against time to rescue "victims" from vehicles and attend to bystanders affected by the chemicals.

These joint exercises build closer working ties between the response forces in both countries. Existing gaps in operations are also looked into and bridged as a result of the collaboration. Together with the emergency response plan, the exercises ensure that forces on both sides of the Tuas Second Crossing are always on the ready for any incidents involving hazardous chemicals.







Participants in a display of solidarity after the exercise in 2011.

From X-Rays To Laser Pointers

Radiation protection in Singapore began when the Radiation Protection Inspectorate (RPI) was formed in 1972, under MOH. The following year, the Radiation Protection Act was passed. The responsibilities for radiation protection and nuclear science were consolidated under NEA in July 2007. Today, NEA discharges its duties to ensure safe and secure use of radioactive materials as well as ionising and non-ionising irradiating devices through a system of licensing, inspection and enforcement.



Thermoluminescent Dosimeter

RADIATION UNDER SCRUTINY

The next time you visit an x-ray room, see if you can spy the Thermoluminescent Dosimeter (TLD) worn by the operators of the machine. This gadget, distributed by NEA to all radiation workers, clocks the amount of radiation they are exposed to. The TLDs are analysed periodically by NEA to ascertain that the annual radiation dosage of each worker does not exceed 20 millisieverts (mSv) per year, the safety limit.

This is just part of the work NEA does to regulate radiation in Singapore. From x-ray machines and laser pointers to microwave ovens and smoke detectors, the agency controls just about any radiation-emitting equipment and substances that can potentially harm humans. Radiation is commonly classified into two forms – ionising and non-ionising. Both can cause harm to life and the environment, depending on the dosage. Ionising radiation also has the energy to split atoms and molecules in its path into ions, and can damage DNA. Its effects on living organisms are cumulative. The uses of radiation, such as in nuclear medicine, radiotherapy, industrial applications and household appliances, however, mean that irradiating equipment and substances are very much a part of our surroundings.

The fact that such controls were established early on gave Singapore an advantage, as there have been relatively few cases of irradiating equipment and substances found without documentation of their origins. This is particularly important as Singapore does not have the capacity to treat radioactive waste, and such equipment and substances have to be re-exported to their countries of origin after their useful lifespan ends.

Part of the challenge of NEA's work is to keep up with advancements in nuclear science and technology, and the growing applications of radiation. In 2008, the agency issued a total of 35,481 licences. By 2010, the number had increased to 37,170. As the medical and industrial sectors expand, so do the scale and complexity of NEA's duty to protect Singapore from the potential misuse and abuse of radiation.

Altogether, NEA administers 17 types of licences to manage the import, export, use, ownership and transport of radioactive materials or irradiating apparatus. Radiation workers have to sit for a test to make sure that their knowledge and skills are sound before they are licensed. Inspections are conducted before and after a licence is granted to ensure that the right conditions are in place. For example, the radiation level outside rooms housing irradiating equipment should not exceed 10 µSv/hr. If the premises or activities are found to be unsafe, the licensee will be warned and required to take mitigatory measures, or even face prosecution.

Opposite: Inspecting to see if radiation levels emitted by the angiography equipment are within permissible limits.



Part of the challenge of NEA's work is to keep up with advancements in nuclear science and technology, and the growing applications of radiation. As the medical and industrial sectors expand, so do the scale and complexity of NEA's duty to protect Singapore from the potential misuse and abuse of radiation.



SHOULD A "DIRTY BOMB" FALL

It is a thought that hardly occurs to the man on the street – what if Singapore were to be hit by a "dirty bomb", or radiological dispersal device, one day?

At the Centre for Radiation Protection and Nuclear Science (CRPNS) in NEA, imagining such scenarios and planning for them are part of the job. As with any other industrialised country, there are facilities in Singapore with equipment and materials that emit radiation. Nuclear-powered naval ships visit the island from time to time. Moreover, a number of neighbouring countries have announced plans to build nuclear power plants in the near future.

Planning for radiological and nuclear crises is not the work of just one agency. Rather, it is a coordinated effort which involves government agencies such as SCDF, MOH, Agri-Food and Veterinary Authority (AVA) and the Ministry of Foreign Affairs (MFA). CRPNS, with its experience and knowledge, brings to the table technical and scientific expertise. In the event of a crisis, its scientists and technical officers will undertake critical tasks to ensure the public's safety, such as monitoring radiation levels, deciding on the extent of the warm zone (zone within which the threat still exists but is not immediate) and testing soil samples for ground contamination.

Readiness goes beyond planning on paper. Joint emergency exercises are conducted together with SCDF. Scenarios involving nuclear-powered warships or radiological dispersal devices are simulated. The centre also works with MFA to participate in regional nuclear safety co-operation.

Having procedures in place does not mean that the work is done. At the end of the day, it is access to relevant information and timely analysis which hold the key to an adequate emergency response. As Ang Kok Kiat, Director of CRPNS puts it, "To respond to such threats, we need to know what nuclear technologies are out there and their inherent safety problems, and the consequences should incidents happen. We must be able to assess the impact from the release of radioactive materials – what will be released, how much will be released and for how long. And meteorological data to tell us whether the materials will come to Singapore, how long it would take, and how long it would last. These are things that are critical in planning for an emergency response."

AN EXERCISE IN VIGILANCE

In March 2011, the tsunami triggered by an earthquake off Japan's coast led to equipment failure and release of radioactive materials at Fukushima Daiichi Nuclear Power Plant, resulting in a nuclear crisis.

Even though the incident took place 5,500 km away, the Singapore government decided not to take a chance. It mobilised relevant agencies to prepare contingency plans and launched responses appropriate to the level of danger presented by the situation.

NEA played a crucial role in assessing the threats faced by Singapore. The Meteorological Service Singapore provided weather data to project if the radioactive plume released would reach Singapore. Using the data, CRPNS calculated that the radiation levels would be negligible by the time it arrived.

Despite the findings, the government adopted a vigilant approach. Four ambient radiation monitors were set up by CRPNS in different parts of Singapore to monitor radiation levels. Water sampling was conducted to check for radiation contamination in our coastal areas. The AVA stepped up its screening of food imported from Japan and CRPNS pitched in to help.

To assure the public of Singapore's safety, readings of radiation levels were published daily on NEA's website. The government released press statements and set up a dedicated website for Singaporeans to keep track of the situation. To allay concerns that radioactive materials could be released when contaminated produce from Japan was incinerated, some four tonnes of food were double-bagged and secured in durable plastic bags before they were carefully disposed of in Semakau Landfill, even though contamination was low and incineration was acceptable under international guidelines.

At the height of the crisis in March and April, NEA's Ops Room, where emergency response operations are coordinated during a crisis, was activated to function 24/7 and daily situation reports were prepared. However, as the crisis began to stabilise in Japan, and readings in Singapore showed constant low background radiation levels of 0.08 – 0.1 μSv/hr, the operation was eventually scaled down by early May 2011.





WHAT TEXTBOOKS DON'T TEACH

"Radiation, what does that have to do with the environment?" That is the most common response when Dr Phua Tan Tee tells people what he does.

Dr Phua, one of the few radiation scientists in Singapore, has been in the line of radiation protection and safety since 1983, when he joined the former Radiation Protection Inspectorate. Today, he heads the Ionising Radiation Control Unit in CRPNS, where much of his work revolves around leading his team to license, inspect and monitor irradiating equipment and materials, and to provide scientific expertise and advice where needed.

Having the right qualifications, however, is not enough for a scientist working here. Effective communication and knowing how to handle human relationships are highly prized skills as well. "The textbooks and teachers can only give you this much. Our knowledge comes from on-the-job training," shares Dr Phua. "Experience on the ground, going out to do inspections, meeting the licensees face-to-face, understanding what they are doing, their environment, the materials they are using, their difficulties and what you are licensing for; that is important. Anyone who joins us, if he doesn't like site work or cannot handle human relationships, he may not be able to stay on."

One of the most striking things about Dr Phua is his ability to sidestep scientific jargon and talk about radiation science in layman's terms. Which is necessary in his job, where he has to work with people from all walks of life, from technicians to doctors, answering their queries and convincing licensees and users of irradiating equipment and substances that NEA's regulations are reasonable. During the Fukushima crisis, the number of public queries relating to radiation spiked. "If the car is imported from Japan, is it ok? Can I mix my daughter's laundry with the rest of our laundry if she just came back from Japan? Are appliances and electronic parts from Japan safe for use?" Dr Phua elaborates, "These are just some of the queries we received and we need to assure the callers. Because there is a lot of misconception about radiation safety and people are very influenced by the movies they watch! So in this line, it is important to have good techniques in communication, to use the right words, the right tone. This can't be taught. Sometimes we even have to go down to the site to assure and counsel those who were excessively worried about radiation in their vicinity."

Like many of his colleagues in NEA, Dr Phua finds his work meaningful. "If there is no control over radioactive materials or equipment, the issue of safety will arise – safety for the user, and for the public," he says. "And on the medical side, safety for the patients as well. And it's quite interesting work! If you love your job and you love to share knowledge, it is enjoyable, sharing information and to being able to assure others that they are safe. That is the most satisfying part."

Trashing Waste For Clean Land

Dealing with waste is a serious affair here. Besides having to dispose of more and more waste every year, this waste has also got to be treated quickly before it becomes a public health and environmental hazard in our hot and humid climate. Waste management in Singapore relies on a few core pillars. Enforcement, investment in infrastructure, innovation and promoting the 3Rs – Reduce, Reuse, Recycle.

CRACKING DOWN ON DUMPING!

Dumping of untreated waste, especially from industrial sources, leads to land pollution and may even cause water pollution if it takes place near waterways. The way the authorities, first ENV, then NEA, cracked down on illegal dumping is an example of how legislation, monitoring, enforcement and education come together in environmental protection.

The usual culprits are earth-moving contractors and general waste collectors. Generally, they do it to avoid paying fees at incineration plants. To escape detection, they often dump at out-of-the-way spots, such as Lim Chu Kang, Tampines, Punggol and Tuas.

Under the Environmental Public Health Act, trashing waste in public and unauthorised places is strictly prohibited. Offenders face fines, jail terms and confiscation of vehicles used to carry out dumping.

The efforts to clamp down on illegal dumping started with toughening the penalties for the act. In 1999, the fine for the first offence was raised from $10,000 to $50,000. This alone, however, could not deter would-be offenders as long as they believed they could get away with it. Effective enforcement was necessary. Surveillance at hotspots was carried out to catch culprits in action. Earth bunds, fencing and bollards were put up to prevent easy access to a convenient dumping ground.

To make the effort even more effective, the public was encouraged to report on illegal dumping. Offenders who were caught also received attention in the press to highlight the stepped-up enforcement. With all these measures, ENV, and later NEA, successfully brought about a sharp decline in such activities. In 1999, there were 190 cases of illegal dumping recorded in Singapore. By 2010, the number had dropped significantly to 39.

ENERGY FROM GARBAGE

The story of how waste-to-energy (WTE) incineration plants became an integral part of our solid waste management strategy illustrates Singapore's willingness to invest in technology for the environment. WTE incineration may seem like an easy choice to make. However, in the early 1970s, it was anything but that. Dumping rubbish directly in landfills was viewed as the most natural course. There were relatively few countries using incineration to dispose of waste, save for some parts of Europe and land-scarce countries such as Japan. Fears of air pollution caused by incineration were widespread, and some developed nations even balked at commissioning incineration plants on a large scale. On top of that, WTE incineration plants were expensive, usually costing more than $100 million each. So when Singapore decided in 1973 to invest in the first WTE incineration plant in Asia outside of Japan, it was a bold move.

Opposite: Refuse bunker at Tuas South Incineration Plant.



In 1970, some 1,200 tonnes of rubbish were disposed of per day. By 2011, the figure had risen to 7,830 tonnes every day.

There were obvious advantages, of course. The volume of waste could be reduced by an amazing 90%, thus saving landfill space. Heat, the by-product of incineration, could be harnessed to produce electricity. An incineration plant took up much less space than a landfill and could process more waste. Scrap iron could also be recovered from the waste. To overcome the concern of air pollution, flue gas cleansing treatment and filter treatment systems were installed to remove acidic gases, dust, and other contaminants.

In 1979, Singapore's first WTE incineration plant, located at Ulu Pandan, was completed at a cost of $130 million. The second WTE incineration plant was commissioned in 1986 in Tuas. Senoko Incineration Plant came about in 1992.

Singapore's fourth and largest WTE incineration plant was commissioned in 2000 at a cost of $890 million. Sited on a 10.5 ha plot of reclaimed land, Tuas South Incineration Plant is one of the largest of its kind in the region in terms of capacity. The plant can burn 3000 tonnes of garbage every day. Industrial and domestic waste, contraband goods, ballot papers and decommissioned currency notes – these are just some of what pass through the plant's furnaces.

An average of 800 refuse trucks roll into the plant per day. They tip their rubbish into a bunker, and are weighed before and after to calculate disposal fees. In the bunkers, air pressure is kept below atmospheric pressure to contain the stench. Cranes mix the rubbish to make sure that the composition is even before feeding it to six incinerators. Advanced combustion control systems regulate the combustion rate so that the refuse can be completely burnt.

The gas produced passes through a precipitator where dust particles, even those smaller than 1 micron, are removed. Next, it goes through a dry scrubber system and a catalytic fabric filter system to neutralise its acid contents and remove dust and other contaminants before exiting through two 150 m tall chimneys. The quality of the flue gas is monitored for pollution control. The resultant ash and slag are transported to pits and electro-magnetic separators to pick out ferrous scrap metal for recycling. The remainder will be sent to Semakau Landfill.

The heat produced is used to generate steam in boilers, which drives generators to produce 600kWh of electricity per tonne of refuse. In total, 20% of the power goes to operating the plant and 80% is sold to the National Electricity Market of Singapore. Exhaust steam is cooled and pumped back into the boilers for re-use.

If there is anything amiss with the equipment, alarms will go off. These alarms work on parameters that are customised and indicate different levels of urgency. Operators can then prioritise their tasks and take pre-emptive action.

Today, Singapore is one of the few countries in Asia to use incineration of refuse to manage waste. Incineration of waste continues to be a controversial topic for reasons such as air pollution and toxin levels in incineration ash. As technology progresses, however, pollution control methods have improved. Emissions by incineration plants are monitored to make sure they are below safety limits. In addition, Singapore's system of separating hazardous waste from incinerable waste and sanitary disposal of ash at Semakau Landfill ensure that the fears of pollution are not realised here.





To keep the plant going, five operations teams work round the clock in three shifts. There are checklists for fabric filters, boilers, generators, turbines – every equipment and process in the plant.

KEEP IT BURNING

"We are the last line of defence. The other incineration plants are smaller. And once their bunkers are full, they will divert the refuse vehicles to us, and we have nowhere else to divert to. So we have got to take in whatever that comes in. We are like their big brother," explains Mohamad Ghazali Bin Ismail, Senior Operations Manager at Tuas South Incineration Plant (TSIP).

Meet the people behind TSIP. These engineers and technicians are a rare breed in Singapore. As Joseph Boey, Senior Manager, says, "Put it this way, there are not too many of us around because there are only four such plants in Singapore." These people keep Singapore's largest incineration plant going, 24/7, 365 days a year. They manage fires that reach temperatures of 1,000°C, so as to combust the tonnes of rubbish delivered to the plant every day. Altogether, over 140 of them work to manage the plant.

NO DOWNTIME!

There are two main groups of people who keep the plant going – the operations team and the maintenance force. One manages the day-to-day running and the other looks after the equipment.

Downtime is the bane of workers here – to be avoided at all costs. If any of the incinerators is down, refuse will start piling, and public waste collectors may not have anywhere else to turn to. Sunny Tan, Higher Technical Officer, has this to say, "We cannot afford downtime. We are always fighting for time."

To keep the plant going, five operations teams work round the clock in three shifts. Then there are checklists for fabric filters, boilers, generators, turbines – every equipment and process in the plant. Selvam, Senior Technician, explains, "The checklists ensure that there will always be men on the ground. A pipe may have a pinhole leak, but the system will show no change in the water pressure. So water will be wasted, and the small leak may lead to something more serious. Or the equipment may appear to be working on the system, but locally, a part may be out of alignment. If we don't correct it, the part may give way! So we do the checks two or three times every shift."

The operations team keep an eye on the refuse coming in as well. Bulky waste, for example, can bring operations to a halt by jamming the incinerator or even damaging the machinery. Four high capacity rotary bulky waste crushers are installed to deal with such garbage. Smouldering refuse and hazardous chemicals are to be watched out for too. Sometimes, waste collectors bring in refuse that may cause fires in the bunkers, which in turn may lead to a plant shutdown. And despite strict regulations banning the disposal of hazardous materials at incineration plants, such substances still end up here now and then.

"We do random checking. Every day, we will check about 30 to 40 lorries. One lorry takes about 10 minutes. If there is rubbish that is not supposed to be there, we will scoop it up and put it back into the vehicle. And we will warn the waste collector," says Mohamad Ghazali. Operations Shift Supervisor, Low Eng Hai, quipped, "Especially during the seventh month or Qing Ming, when people burn offerings. When we see vehicles coming in with smoke coming out, we will tell them to unload their refuse. And if it is on fire, we will spray water. Bunkers catching fire is very rare and we have zonal sprays to put them out. If there is a fire, it's an emergency! Prevention is better than cure!"

The bunkers at TSIP are able to hold up to four days of trash, but forward planning is still essential. During festive seasons and mass events such as Christmas and the Formula One night race, the volume of waste may spike by 30% or more. The operations team has to make sure that bunker levels are low during such times.

Opposite: Inspection and maintenance works inside the furnace during an overhaul.



Every year, each of the six incinerators at TSIP is shut down for a month for an overhaul. The overhaul has to be timed during periods when waste volumes are low. The incinerator and all its parts will be cleaned, thoroughly checked, maintained, repaired and tested before it is allowed to start again.

To keep the equipment at the plant going smoothly, maintenance and incinerator overhauls have to be carried out. The equipment is maintained regularly. When issues crop up, the maintenance crew, which is on call 24/7, will look into them before they escalate. Dennis Lee, Technical Officer, says, "Our work involves maintaining an incineration plant and a power generator. The work here is very complex, from the equipment to the treatment process."

Every year, each of the six incinerators at TSIP is shut down for a month for an overhaul. The incinerator will be cleaned, checked, maintained and tested before it is allowed to start again. Technicians such as Muhammad Khairozi Bin Hassan will inspect each part methodically. He says, "For example, we use ultrasonic equipment to test if the boiler tubes are too thin. Too thin, they will burst. We have to check the inside of the furnace, which is 40 m high. We take all the readings, and the technical officers will analyse to see if repairs are necessary." All things electrical are covered by an electrical maintenance team.

Before the incinerator can start up again, the pressure test, which lasts a day, will be conducted in the presence of an officer from the Ministry of Manpower. The incinerator's boiler is filled with water and the pressure is increased steadily until it reaches 40 barG. Once it is demonstrated that there are no leaks, the incinerator is ready to burn.

AT THE END OF THE DAY

The work at TSIP is demanding, challenging and unpleasant even. Hardship is a constant. The people have to work in adverse conditions of dust, odours, extreme heat and noise. Workplace safety is deeply ingrained, because, as Low points out, "It's a matter of life and death."

Safety procedures, meetings at the start of the day to reiterate safety concerns, personal protection equipment, a website and courses dedicated to safety, drills on chemical leaks and fires – all these go to creating a strong safety culture. This can be seen in operations such as overhauls, when workers have to operate in confined spaces, facing potential threats such as carbon monoxide. Trained assessors will check at least every eight hours if the area is safe to work in. Workers are also equipped with gas meters to alert them to any dangers. Incompatible tasks, such as painting and hot work, are not allowed to take place simultaneously.

Despite all these, TSIP's people are proud of their work. Tan Kee Pin, Principal Manager, highlights, "It's more complicated to maintain an incineration plant, because there is a power plant too. People who join us have to start from zero to understand the process. Some people can't take this kind of environment and can't stay on. But those who stay are those who love the work and challenges despite the hazards."

What about the smell then? "Immune!" laughs Low, "Definitely I'm very proud of my work. It's a unique job. Those who love working with machines will love working here."

And contrary to expectations, people around them are fascinated rather than put off by their work. Tan Kee Pin divulges, "Because there are only four plants in Singapore, a lot of people don't know what we do. When I tell them that we burn rubbish and from the heat, we produce electricity, they are actually very fascinated."

For Joseph Boey, "I burn garbage" is a handy line to explain his work. Joseph goes, "My seven year old son asked me what my job is. And I said, 'I burn garbage'. He replied, 'You burn garbage, how come I don't smell the garbage when I sit beside you?' And this is the amazing part of my job. I'm in this industry and I face tonnes and tonnes of garbage every day. But technology has progressed to such an extent that we don't have to be associated with rubbish in a negative way anymore. This job has got no lack of challenges. The plant takes care of the majority of the waste island-wide. I have to do my part to make sure that this plant serves Singapore. That in itself is a meaningful responsibility."



Top: Checking electrical parts in the plant.
Opposite from top: Part and parcel of work - inspecting the turbine-generator; Making sure that the stokers are in tip-top condition.





THE MATHEMATICS OF TUAS SOUTH INCINERATION PLANT

Incineration capacity
3,000 tonnes a day

Catalytic bag filters comprise
2,240 fabric bags

Temperature at which refuse is burned
900 - 1,000°C

Steam generation per boiler
370°C (35 barG)

105 tonnes per hour

Power generation capacity
80 MW
10.5 kV generator voltage

24 refuse discharge bays

8 Weighbridges

6
Ten-compartment catalytic bag filters
Two-zone electrostatic precipitators
Stoker/furnace units with integral boilers

4
Bulky waste rotary shears
Refuse cranes of 10 m^3 each

3 Overhead wire rope operated ash cranes of 3.5 m^3 each

2
Air-cooled condenser systems with 12 fans each
Bulky waste cranes
Concrete chimneys 150 m in height
Condensing steam turbines coupled to generators
Lime silos and reaction product silos
Magnetic separators to each incinerator unit
Pusher-type hydraulic slag extractors to each incinerator unit
Rain water buffer basins
Separate refuse storage bunkers
Vibrating conveyors to each incinerator unit



The heat produced by the burning refuse does not go to waste. It is used to generate steam in boilers, which drives two steam turbines coupled to generators to produce electricity. In total, 20% of the power goes to operating the plant, and 80% is sold to the National Electricity Market of Singapore. The exhaust steam is then cooled and pumped back into the boilers to be used again.

INNOVATION FOR WASTE

Waste management is complex. Collecting it in a way that does not create nuisances for the public, treating it, making sure that there is a good disposal system – such operations are all the more challenging in this high-density, compact city. By using innovative engineering and technology, NEA manages to improve many aspects of solid waste management.

Public waste collection generally relies on a system of garbage trucks making the rounds to every block and building and emptying their chutes. During these collection times, noise and smells cannot be avoided. Progressively, NEA has implemented a system whereby these issues no longer plague waste collection.

The Pneumatic Refuse Conveyance System (PRCS) uses air suction and a network of pipes to suck rubbish from buildings to a central location for collection. Such a system has several advantages. The odour of garbage during collection, traffic noise, scavenging by rodents and leaking chutes — all these issues become irrelevant as collection is done in a closed system. The time and manpower required for collection are cut down significantly too. Today, PRCS is used in some 60 residential and commercial developments in Singapore, including hospitals, shopping malls, residences and hotels.

Another innovation explored by NEA is the use of incineration bottom ash (IBA) to lay roads, a method used in countries such as Netherlands and Taiwan. IBA is the matter left behind after waste is burned. In 2010 alone, some 405,000 tonnes of IBA were sent to the only landfill in Singapore, Semakau Landfill.

To reduce this volume, NEA embarked on an experiment between 2001 and 2003. Working with the Land Transport Authority (LTA), PUB and a private company, NEA conducted a trial to pave 150 m of road along Jalan Buroh with 1,200 tonnes of treated IBA. The process was tricky as IBA contains traces of heavy metals and incompletely burnt materials. However, the experiment was a resounding success. The quality of road was comparable to those built using conventional materials. Groundwater and storm water run-off from the section did not show signs of contamination. Following that, another two sections along Tampines Road were laid with IBA by LTA. The sections were tested for levels of hazardous substances such as carbon, dioxins, phosphate etc. The conclusion? IBA-built roads are good to go.



Former Minister for the Environment Lim Swee Say laying the road with IBA.

Such application of IBA, however, is yet to be commercially viable in Singapore, for reasons such as the high cost and uncertainty over market demand for the material. Nonetheless, NEA remains open to the idea that this method may be used one day, while continuing to look out for more solutions to tackle the waste issue.






Wildlife that thrive on and around Semakau Landfill include (*from left*) the grey heron, jumping spider, nudibranch, feather star.

HAVEN ON A DUMP

Keep still, and you may very well catch a glimpse of endangered birds such as the Great-Billed Heron and Malaysian Plover. The waters teem with barracuda, trevally and milkfish. During low tide, you may encounter the Giant Carpet Anemone, sea urchins, sickle seagrass, nudibranchs and the impressive Knobbly Sea Star. Mangroves line the edge of the mudflats, offering an oasis for intertidal wildlife.

Sounds like a nature park, but it is actually a garbage dump. Pulau Semakau, 350 ha in size and capable of holding 63 million m^3 of garbage, is Singapore's only landfill. What is unusual about Pulau Semakau is that it is the first-of-its-kind landfill in the world that demonstrates how natural habitats can co-exist with waste.

Managed by NEA today, Semakau Landfill is where all of Singapore's incinerated trash and non-incinerable waste go. Barges regularly ship refuse from Tuas to Semakau Landfill. The rubbish is then transported to a series of cells that serve as landfill. These cells, located in an artificial lagoon, are filled with seawater until they are to be used. To get a cell ready for waste-tipping, water is first pumped out of the cell, which is lined with impermeable geomembrane to prevent waste from seeping into the sea. When a cell is used up, it will be covered with earth. NEA regularly samples the water surrounding the landfill for contamination.

This feat of engineering began in the 1990s, when it was clear that Singapore's last landfill at Lorong Halus could not keep up with the island's waste management requirements. After considering future development needs on the main island, the government decided to build the landfill offshore. In a stroke of creativity, engineers at ENV (present-day MEWR) proposed to turn the sea space between Pulau Sakeng and Pulau Semakau into a landfill.

However, the ministry had to first assess the feasibility of the proposal and the landfill's potential impact on the marine environment. As this would be the first landfill of its kind in the world, it sent engineers to Japan to learn about near-shore landfills and to the United States to look at barging systems for offshore facilities. A consultant was engaged to study the viability of the idea and whether the impact on the marine environment could be mitigated. The conclusion was that the idea could be realised. Approval was given by the parliament in 1994 and construction began in 1995.

That was not all. Biologists were consulted and measures taken to minimise the impact on the ecosystems of Pulau Semakau and Pulau Sakeng. About 400,000 mangrove saplings were planted to replace those destroyed in the construction process. New plots of mudflats were created on the fringes of Pulau Semakau before the replanting could take place. Artificial creeks were built to ensure that seawater could circulate to the saplings. To protect the corals around the island, the engineers used fine mesh silt screens near the work areas to reduce the impact of sediment. Four years after the project began, Semakau Landfill officially opened on 1 April 1999. Completed at a cost of $610 million, it is expected to meet Singapore's needs for landfill space beyond 2045.

Today, the first thing that is noticeably absent when you step on Pulau Semakau is the smell of garbage, usually the signature of a landfill. Instead, you can sight over 80 species of birds, 8 species of seagrasses, more than 40 species of butterflies and a host of corals, anemones and marine life on this unlikely nature haven, found, of all places, on a rubbish dump.

"Some 10 years ago, we were running out of space on the mainland to build any more landfill for our garbage. The engineers in NEA then came up with an innovative solution to build an offshore landfill. It was never tried out in such a big scale anywhere else in the world before, but our engineers persevered for four years with the hard engineering work involving more than 1000 workers to build the rock bund from the bottom of the seabed. They also took care to put in measures to protect the marine ecosystem. That Semakau today is not just a landfill but an island with opportunities for recreation and nature lovers is truly a great testimony to the forward thinking and excellent planning of our engineers."

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011) at the National Engineering Week, 24 October 2005





A PLACE TO BE ENJOYED

The man who pushed for Pulau Semakau to be opened to the public, Dr Yaacob Ibrahim, had the brainwave after his first visit to the island in 2004. "We took the boat from Pasir Panjang," says Dr Yaacob. "If you remember that road, you move away from the landfill station, and then you see the mangroves, you see the greenery. I said, 'This cannot be a landfill, right?'"

The visit was a revelation of sorts. Pulau Semakau was not just any landfill. In the words of the former minister, it had the potential to be a place where "we show Singaporeans that we can balance diversity and environmental management in a very nice way." Dr Yaacob thus took the plunge to open the island to the public. In 2005, NEA approached organisations such as Raffles Museum of Biodiversity Research to look into whether the island could be used for recreational and educational activities. After some studies, it was found that Pulau Semakau offered a rich environment for guided nature walks, bird watching and sport fishing outings.

Nature lovers, anglers, school groups, researchers and just about anyone who was curious started flocking to the island. Its rich biodiversity won converts easily and the story began spreading. International media soon caught onto it. Word of a nature haven on a rubbish dump appeared in the papers, news channels, documentaries, scientific and environmental journals, and along with it, accolades and praise for the innovative solution. Officials from various nations such as Malaysia, the Philippines, New Zealand, Japan and Brunei came to study the possibility of replicating the landfill in their own countries.

Meanwhile, the island has grown to become a favourite recreational spot for Singaporeans. Guided walks are conducted regularly to introduce participants to the fascinating intertidal ecosystem. Bird watchers, anglers and people interested in astronomy find Pulau Semakau one of the best places for their activities. NEA's first environmental charity run, benefitting local environmental NGOs, was also held on Pulau Semakau in 2007, with more than 40 corporations taking part.

To Dr Yaacob, Pulau Semakau remains a special place. He recounts a trip that he once took with his family. "As we were walking out, it was already 7 pm, getting quite dark, and we saw these lights in the trees. Fireflies! We saw fireflies in Semakau! It showed that the water was clean enough for the fireflies to come back. At the end of the day, we managed to bring back the biodiversity. The birds came back, the crabs came back and I thought it was fantastic. It is one of the little areas that reminds you that it is possible for us to create in a sense, a nirvana out of Singapore."



Aerial view of Semakau Landfill, 2011.

WASTE AS RESOURCE

It's time to give waste a new lease of life because it can be a rich resource too.

Resource efficiency is actively promoted by NEA as a strategy to cope with waste. For instance, the agency works closely with recycling companies, the Building and Construction Authority (BCA) and the construction industry to promote the recycling of construction and demolition waste. Today, about 99% of such waste finds new uses. Used copper slag is adapted for grit blasting; steel slag, for the production of road-building materials and used concrete, for foot paths and road curbs. Another exciting project focused on the recycling of ink and toner cartridges. This project, which commenced in 2011, was initiated by manufacturers. NEA supported them by providing advice and facilitating meetings between the manufacturers, recycling companies and the National Library Board, whose libraries housed the drop-off bins for the cartridges.

Recycling went nation-wide in 2001 under ENV. In public areas with high human traffic, recycling bins were placed. Schools became part of the national recycling campaign when NEA started a programme in 2002 to inculcate such habits among students. By 2008, all condominiums and private apartments in Singapore were required to have recycling receptacles within their premises. To drive home the message of "reduce, reuse, recycle" among Singaporeans, NEA kicked off Recycling Day in 2004. This was extended to become Recycling Week in 2010.

The various measures have brought about a gradual improvement in our waste situation over time. In 2001, the national recycling rate was 44%. As of 2011, the recycling rate has reached 59%.

SINGAPORE PACKAGING AGREEMENT

Less of waste means more going for environment sustainability. Today, more than 100 companies and industry associations in Singapore have banded together to reduce packaging waste, thanks to an agreement launched in 2007 by NEA.

The agency began exploring the idea of reducing packaging waste right at source in 2005. After consultations with the industry, it concluded that the best way forward was to facilitate a voluntary agreement among stakeholders. In 2006, NEA formed a task force comprising members from the industry, NGOs and the government to develop the Singapore Packaging Agreement (SPA).

Albert Lim, Chairman of the Packaging Council of Singapore, was invited to co-chair the task force along with NEA and Singapore Environment Council. His expertise and experience were crucial to the development of the SPA.

In 2007, the five-year SPA was finally signed by 32 organisations, including five food and beverage industry associations representing more than 500 companies. The NEA also formed a Governing Board to oversee the implementation of SPA, with representatives from different stakeholders.

The board set to task immediately. A Code of Practice was developed to guide designers and users in addressing environmental concerns in their packaging decisions. Activities were organised. "Every few months, we invite signatories, non-signatories and relevant government agencies for a networking session. We present case studies. Companies can share the problems they face with their waste and others can help, or we can match the businesses for solutions. We also circulate information on what other countries do with regards to packaging waste and invite local and overseas speakers to share their views," says Lim. At these networking sessions, non-signatories were also encouraged to participate in SPA.

Membership began to grow steadily. At first, only food and beverage packaging was covered under SPA. In October 2009, the agreement was extended to include all types of product packaging including detergents and electronics. Commercial building owners, shopping malls and supermarkets also joined the agreement to support the waste reduction drive.

This programme has yielded encouraging results. Within the first year, these organisations had managed to save 850 tonnes of waste collectively. Four years into the agreement, the signatories cumulatively saved more than 7,000 tonnes of waste. In 2008, the 3R Packaging Awards were inaugurated to recognise signatories who had made commendable progress in achieving the agreement's objectives.

A second SPA is to be launched in July 2012. Says Mr Lim, the man who played no small role in making SPA possible, "I have been in the packaging industry for the past 36 years. This is one of the things I want to be involved in purely because packaging has been part of my life. The amount of packaging in Singapore is quite high, so somehow it is also a personal achievement to see the waste being reduced, and the change taking place in the packaging industries now."





3R Packaging Awards Ceremony, 2011.



"It took us about a year to develop SPA. We visited New Zealand to study their packaging agreement and adapted it for Singapore. There were consultations with the industry. One of the biggest challenges was getting signatories on board. We had to talk to the companies and convince them that they could benefit by reducing packaging waste. In the end, they realised that raw material costs were increasing, and the only way to keep packaging costs down was to cut packaging usage or reduce packaging wastage."

— Albert Lim, Chairman, Packaging Council of Singapore and Chairman, SPA Governing Board

"At Tetra Pak, we continuously refine our processes in the manufacturing of beverage cartons. For example, we identified the possibility of reusing trimmed polyethylene from our production line within a closed loop. The material was fed back into the lamination process, thereby reducing waste. This required modifications to our equipment and capability development to ensure that product quality was not compromised. Through the Singapore Packaging Agreement, Tetra Pak was able to share with other signatories our innovations and relevant contacts to facilitate the recycling of composite material containing polyethylene and aluminium, which are also used in other food and beverage packaging. We also work with different stakeholders to make sure beverage cartons are recycled as broadly and effectively as possible. Since 2008, we have been running the nation's largest school recycling competition, to encourage students to recycle used beverage cartons. Today, the Tetra Pak School Recycling Competition has expanded to include 100 schools islandwide."

— Jacques Deschamps, Cluster Leader for Supply Chain Operations in South and South East Asia, Tetra Pak

"Nestle has always embraced efforts towards a sustainable environment. To affirm this commitment to our stakeholders, we readily agreed to be a pioneer signatory to the Singapore Packaging Agreement in 2007. This platform has given us the opportunity to showcase our initiatives towards a sustainable environment, and by doing so, we hope to encourage others to follow suit. We are also kept abreast of what other manufacturing and packaging companies are doing and learning from them as well. By continuously reviewing the design of our packaging and improving our production processes, we have been able to make better and more efficient use of packaging resources. We have saved about 25 tonnes of tin, 67 tonnes of plastics and 27 tonnes of paper every year. Besides walking the talk, we would like to encourage all others to do their part towards a sustainable environment by recycling packaging waste. Hence, we developed a smartphone application, 123Recycle, with Nanyang Polytechnic to help educate consumers on how and where to recycle packaging waste."

— Valerio Nannini, Managing Director, Nestlé Singapore

A Liveable Place

"For me, I think it is important that NEA excels as an effective and efficient regulator. There are three things we want to achieve. Firstly, we want Singapore to have the best environment possible — clean air, clean water and clean land. And also a quieter environment. Secondly, we want to build a technically strong and professional team to ensure operational excellence. And thirdly, we aim to provide excellent service quality to the public and industry."
— Koh Kim Hock, Director-General, Environmental Protection Division, NEA



Of all the days in a year, more than 90% of them are days where Singaporeans enjoy good air quality. The raw water from our catchment areas is safe to be treated into drinking water. The beaches are healthy, and our children may play in the waters without fear. All our inland waters support aquatic life.

In a short space of 10 years, NEA has built on the legacy of the former ENV to improve environmental management and catalyse changes in the way things were done. It has also wrought a closer and more meaningful working relationship between the government and the private sector, as well as raised the quality of the environment in Singapore.

Our environmental performance has also garnered international attention. One such example is Singapore's ranking in the Siemens Asian Green City Index, launched in February 2011. The index was put together by the Economist Intelligence Unit, a business entity within The Economist Group. It measured the environmental performance of major cities in Asia. Among the 22 cities surveyed, Singapore was the only city to receive a ranking in the top band of performance, labelled "Well Above Average". In eight of the categories used to assess the cities' environmental performance, Singapore scored "Well Above Average" for water and waste management and "Above Average" for all the other six categories including energy, land use and transport.

The index report noted, "... it is comprehensive and effective policies that elevate the city to rank well above average overall. A rich city-state, Singapore has access to resources, but unlike other cities in the Index, the government is not split between competing levels of administration. And it has a highly trained civil service, along with a reputation for transparency, which is underlined by Singapore's fourth place in Transparency International's Corruption Perception Index."

The importance of political commitment in bringing about a sustainable urban environment was underscored in the speech given by Mayor of South West District, Dr Amy Khor, at the launch of the index in Singapore. She said, "As you are aware, Singapore is a city with limited natural resources and land space. Since the early days of our nation-building, careful thought has been put into how we manage our scarce resources and achieve economic growth, while maintaining a good quality living environment for our people. Today, Singapore enjoys a reputation as a clean and green garden city, through long-term planning and an integrated approach to urban management. Where we encountered constraints, such as our shortage of water, we turned this into an opportunity and developed NEWater. Likewise, we treat waste as a resource and convert waste into energy. This culture of continuous innovation underpins much of Singapore's efforts to overcome its constraints. Our reputation as a clean and green city is the result of continuous effort and cooperation from all stakeholders from the government, industry and community."

The efforts and hours NEA officers put into safeguarding the environment are well worth it. The reward? A clean, liveable, sustainable place for today, and for Singapore's future generations.



In a short space of 10 years since NEA was formed, it has built on the legacy of the former ENV to improve environmental management and catalysed significant changes in the way things were done. It has also wrought a closer and more meaningful working relationship between the government and the private sector, as well as raised the quality of environment in Singapore.